Macro-Conditional Normalizing Flow for Portfolio Risk Management

Ben Komara
Université Paris Dauphine–PSL

Benjamin Fernandes Neres
Université Paris Dauphine–PSL

Dorian Deilhes
Université Paris Dauphine–PSL

Abstract

We propose a macro-conditional normalizing flow for portfolio risk management. Rather than predicting point returns, we estimate the full joint conditional distribution $p(X_t \mid M_{<t})$ of the daily log-returns of a three-asset portfolio (SPY, TLT, GLD), conditioned on the past 63 trading days of macroeconomic features. A Temporal Fusion Transformer (TFT) encodes the macro history into a regime context vector h_t, which conditions a Masked Autoregressive Flow (MAF) over asset returns. The model is trained by maximising the conditional log-likelihood and evaluated out-of-sample on the 2022–2024 stress period via daily Value-at-Risk (VaR) and Expected Shortfall (ES) estimates, formally validated with Kupiec's Proportion of Failures test. All data transformations enforce strict point-in-time integrity using FRED publication vintages, and scalers are fitted exclusively on the training set. Across 438 out-of-sample test days (2022–2024), all three models fail the Kupiec Proportion of Failures test at the 5% significance level. The Gaussian VaR produces a 2.97% breach rate, the Gradient Boosting quantile baseline 2.51% ($p = 0.008$), and the TFT+MAF 4.34%. However, a six-fold expanding walk-forward cross-validation over 1,131 out-of-sample days (2018–2023) yields a 1.24% aggregate breach rate for the TFT+MAF, passing the Kupiec test ($p = 0.438$). This indicates that the single-window failure is driven by the structurally anomalous 2022 tightening cycle rather than a fundamental deficiency of the architecture, and that periodic retraining on expanding windows restores acceptable tail calibration.

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1 Introduction

1.1 Financial Motivation

Modern financial regulation, codified in the Basel III framework, requires banks and asset managers to quantify portfolio risk through two primary measures: **Value-at-Risk** (VaR) and **Expected Shortfall** (ES). The 99% daily VaR is the loss threshold v such that

$$\Pr\big(r_t^{(p)} < v\big) = 0.01,$$

where $r_t^{(p)}$ denotes the equal-weighted portfolio return on day t. The corresponding Expected Shortfall measures the average loss in the worst 1% of outcomes:

$$\mathrm{ES}_{0.01} = \mathbb{E}\big[r_t^{(p)} \mid r_t^{(p)} \leq \mathrm{VaR}_{0.01}\big].$$

Unlike VaR, ES satisfies subadditivity and is therefore a *coherent* risk measure: the ES of a portfolio cannot exceed the sum of the individual ES values, a property that VaR violates for non-elliptical distributions. Basel III requires banks to report ES at the 97.5% level as the primary capital adequacy metric, making accurate tail modelling a regulatory obligation.

Both measures are defined entirely through the *tail* of the return distribution. Computing them accurately therefore requires a faithful model of the conditional distribution $p(r_t^{(p)} \mid \mathcal{F}_{t-1})$, where $\mathcal{F}_{t-1}$ denotes the publicly available information set at the close of day $t-1$.

1.2 Limitations of Classical VaR Models

The industry standard for over two decades has been the **parametric Gaussian VaR**: fit a multivariate normal distribution to historical returns and extract quantiles analytically. This approach rests on three assumptions that are systematically violated in practice.

1. **Thin tails.** Financial returns exhibit excess kurtosis: extreme losses occur far more frequently than any Gaussian model predicts. A Gaussian distribution assigns near-zero probability to daily equity drawdowns of 5% or more; such events occur multiple times per decade in US equity markets.
2. **Stationarity.** The variance of daily returns is not constant. Calm periods cluster together, as do turbulent ones, a property known as *volatility clustering*. A model estimated on a single $\hat{\sigma}$ over the training window is blind to regime changes.
3. **Constant correlations.** Cross-asset correlations are not stable. In stress periods, correlations among risky assets tend to spike toward one, drastically amplifying portfolio-level tail risk. A static covariance matrix cannot capture this phenomenon.

These three empirical failures jointly motivate replacing the Gaussian assumption with a *flexible, time-varying conditional density*.

1.3 Problem Statement and Contributions

Let $X_t = (r_t^{\mathrm{SPY}}, r_t^{\mathrm{TLT}}, r_t^{\mathrm{GLD}})^\top \in \mathbb{R}^3$ denote the vector of daily log-returns at time t, and let $M_{<t} \in \mathbb{R}^{63\times 8}$ denote the matrix of macro and market-regime features over the 63 trading days preceding t. Our goal is to learn

$$p_\theta(X_t \mid M_{<t}),$$

a parametric conditional density that accurately represents the tail behaviour of joint asset returns given the prevailing macroeconomic regime.

This project makes three contributions:

1. **Point-in-time data engineering.** We construct a macro feature dataset that strictly respects FRED publication vintages, preventing any look-ahead bias in the feature matrix.
2. **TFT+MAF architecture.** We propose a two-stage deep generative model: a Temporal Fusion Transformer (TFT) [**?**] that compresses the macro history into a regime vector h_t, conditioning a Masked Autoregressive Flow (MAF) [**?**] over joint asset returns.
3. **Statistical validation.** All VaR estimates are formally tested out-of-sample via Kupiec's Proportion of Failures (POF) test [**?**], providing a rigorous calibration assessment.

2 Data

2.1 Asset Universe and Market Data

We construct a three-asset portfolio comprising:

- **SPY**: SPDR S&P 500 ETF, exposure to US large-cap equities.
- **TLT**: iShares 20+ Year Treasury Bond ETF, long-duration US government bond exposure.
- **GLD**: SPDR Gold Shares ETF, physical gold, a classical safe-haven asset.

These three assets span fundamentally different risk factors (growth, interest rates, inflation hedge) and form a canonical diversified institutional portfolio. Daily adjusted close prices are sourced from Yahoo Finance, and log-returns are computed as $r_t = \log(P_t/P_{t-1})$. Log-returns are preferred for their time-additivity: $\log(P_T/P_0) = \sum_{t=1}^{T} r_t$.

2.2 Macroeconomic Features

The feature matrix $M_{<t}$ combines five macroeconomic indicators with three realized-volatility proxies, for $F = 8$ features in total. All macro series are downloaded from FRED with full vintage history; the VIX is sourced from Yahoo Finance. Table 1 summarises the feature construction.

Table 1: Macro and market-regime features ($F = 8$). All FRED series are downloaded with `realtime_start` dates for point-in-time alignment.

Feature	Source	Raw series	Transformation
CPI	FRED (CPIAUCSL)	Monthly level	Year-over-year % change
NFP	FRED (PAYEMS)	Monthly level	Monthly first-difference
FedFundsRate	FRED (DFF)	Daily rate	Monthly first-difference
HYSpread	FRED (BAMLH0A0HYM2)	Daily spread	Level (mean-reverting)
VIX	Yahoo Finance (ˆVIX)	Daily index	Level (mean-reverting)
SPY_RealVol21d	Computed	Daily returns	$\sqrt{252}\,\mathrm{Std}(r_{t-20:t})$
TLT_RealVol21d	Computed	Daily returns	$\sqrt{252}\,\mathrm{Std}(r_{t-20:t})$
GLD_RealVol21d	Computed	Daily returns	$\sqrt{252}\,\mathrm{Std}(r_{t-20:t})$

The stationarity transformations follow standard practice. CPI is converted to a year-over-year inflation rate to remove both the long-run price-level trend and seasonal patterns:

$$\mathrm{CPI_YoY}_t = 100 \cdot \Big(\frac{\mathrm{CPI}_t}{\mathrm{CPI}_{t-12}} - 1\Big).$$

NFP is first-differenced to obtain monthly job additions/losses (the relevant economic signal), and the Fed Funds Rate is resampled to monthly frequency and differenced to emphasize policy changes rather than the level of rates. HYSpread and VIX are kept in levels, as both are near-stationary stress gauges that revert toward long-run means.

2.3 Point-in-Time Alignment

The core data engineering contribution is the **strict point-in-time (PIT) alignment** of all macro series. Economic data is subject to publication lags:

- CPI for March is published on approximately April 12, it is invisible before that date.
- NFP for month m is released on the first Friday of month $m + 1$.
- FRED's vintage system records, for each observation, the `realtime_start` field: the first date on which that particular value became publicly available.

For each trading day t, we select the most recently published value of each macro series:

$$v_t = \max\big\{s \mid \texttt{realtime_start}(s) \le t\big\}.$$

This is implemented via `pd.merge_asof(..., direction='backward')` on the FRED vintage DataFrame. After the merge, macro values are forward-filled to the next publication date, reflecting the information actually available to a market participant on each day.

This PIT pipeline prevents *look-ahead bias*: the systematic error in which a model is trained on information that was not yet public on the training dates. Ignoring publication lags produces artificially inflated in-sample performance and unreliable out-of-sample risk estimates.

2.4 Rolling Realized Volatility

For each asset $a \in \{\mathrm{SPY}, \mathrm{TLT}, \mathrm{GLD}\}$, we compute the 21-trading-day rolling annualised realized volatility:

$$\hat{\sigma}_t^{(a)} = \sqrt{252} \cdot \mathrm{Std}\big(r_{t-20}^{(a)}, \ldots, r_t^{(a)}\big).$$

These features provide high-frequency regime signals at a daily timescale, complementing the lower-frequency macro indicators and directly capturing volatility clustering as an explicit model input.

2.5 Exploratory Data Analysis

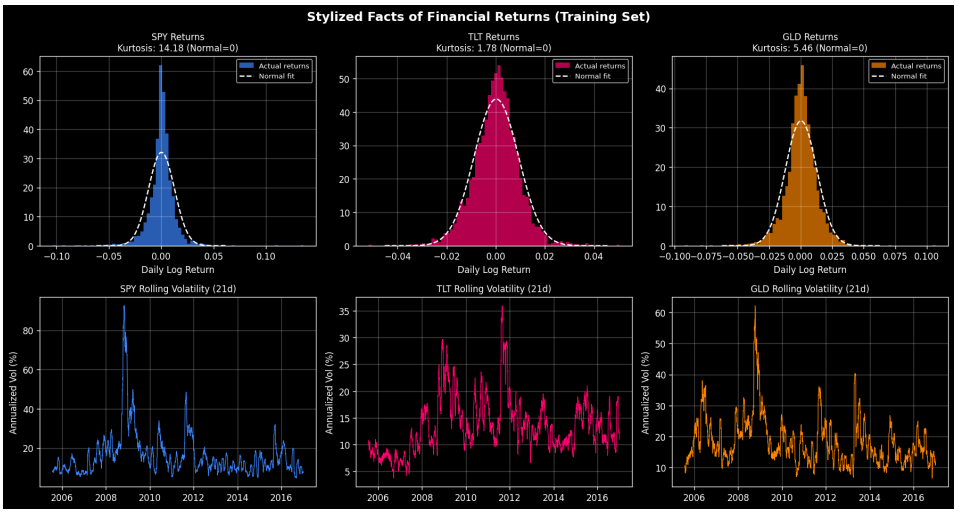

Figure 1: Top row: histograms of daily log-returns for SPY, TLT, and GLD over the training period (2005–2016) with overlaid Gaussian fits. All three assets exhibit excess kurtosis (heavier tails than the Gaussian). Bottom row: 21-day rolling annualised volatility, demonstrating pronounced volatility clustering. These two stylized facts directly motivate a heavy-tailed, regime-adaptive density model.

Figure 1 documents two fundamental empirical properties of financial returns. First, the return histograms reveal pronounced **excess kurtosis**: the empirical tails are far heavier than any Gaussian fit predicts. This implies that Gaussian VaR will systematically underestimate the frequency of large losses. Second, rolling volatility exhibits clear **volatility clustering**: calm months persist, as do turbulent ones. A constant $\hat{\sigma}$ is therefore an inadequate risk summary.

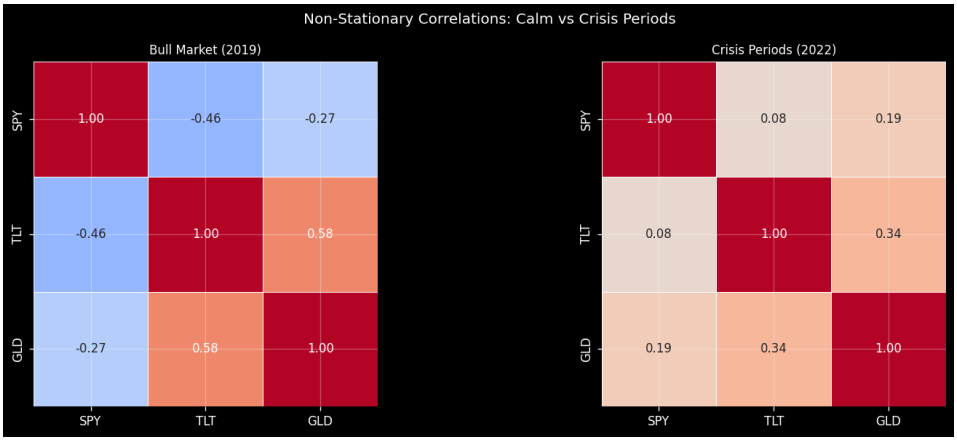

Figure 2: Asset return correlation matrices during a calm period (2019) and a stress period (2022, the aggressive Fed tightening cycle). The typical negative SPY–TLT correlation observed during calm markets weakens or reverses under the 2022 rates shock, amplifying portfolio-level tail losses. This correlation instability justifies modeling the *joint* return distribution rather than three independent marginals.

Figure 2 highlights **correlation instability**. During the calm year 2019, SPY and TLT exhibit a negative correlation consistent with the flight-to-quality dynamic: when equities sell off, investors rotate into long-duration Treasuries. During the 2022 stress episode, characterised by aggressive monetary tightening, this negative correlation weakens substantially as both equities and bonds decline together under rising rates, reducing the portfolio's natural diversification precisely when it is most needed. A static covariance matrix cannot adapt to this dynamic.

Taken together, these three stylized facts, fat tails, volatility clustering, and correlation instability, constitute the full empirical motivation for a conditional, heavy-tailed, joint density model.

2.6 Dataset Construction and Splits

The master dataset spans 2005-06-28 to 2023-12-29, covering 4,659 trading days and 11 columns after PIT alignment and NaN removal (initial rows are dropped due to warm-up requirements: 12 months for the CPI YoY transformation, 21 trading days for the rolling-volatility features, and the Fed Funds Rate vintage availability starting mid-2005). After applying a 63-day sliding window, the dataset yields the shapes in Table 2.

Table 2: Chronological data splits after 63-day sequence windowing ($T = 63$, $F = 8$, $D = 3$).

Split	Period	Sequences	Tensor shape
Train	2005–2016	2836	$(2836,\ 63,\ 8)$
Validation	2017–2021	1196	$(1196,\ 63,\ 8)$
Test	2022–2024	438	$(438,\ 63,\ 8)$

The split is strictly chronological: no future information leaks into the training or validation periods. Each sample $(M_{<t}, X_t)$ pairs a (63×8) macro history window with the target return vector on the day immediately following the window. The training DataLoader shuffles the order of windows (not the internal chronological order within each window) to prevent the optimizer from being biased toward the most recent regime observed within the training period.

2.7 Scaling

Two scalers are fitted *exclusively* on the training set:

- **RobustScaler** for macro features: $z = (x - \text{median})/\text{IQR}$, where $\text{IQR} = Q_{0.75} - Q_{0.25}$. This scaler is robust to the extreme spikes in VIX and HYSpread that occur during crises, preventing a handful of outlier observations from dominating the normalization.
- **StandardScaler** for returns: $z = (x - \hat{\mu})/\hat{\sigma}$. Returns are already approximately centered near zero with bounded variance, making z-scoring appropriate.

Both scalers are applied identically to validation and test data without refitting. Fitting on test data would constitute leakage, inflating the apparent quality of the scaling and corrupting the out-of-sample evaluation.

3 Model Architecture

3.1 Conditional Density Estimation via Normalizing Flows

A **normalizing flow** defines a probability density over $\mathbb{R}^D$ by constructing an invertible, differentiable mapping $f_\theta : \mathbb{R}^D \to \mathbb{R}^D$ that transforms a simple base distribution p_Z into a complex target density p_X. If $z = f_\theta(x; h)$, the change-of-variables formula gives:

$$\log p_X(x \mid h) = \log p_Z\big(f_\theta(x; h)\big) + \log\left|\det\left(\frac{\partial f_\theta}{\partial x}\right)\right|,$$

where h is a conditioning context vector. The term $\log|\det J|$ is the log absolute Jacobian determinant, which accounts for the volume change induced by f_θ. Computing $\det J$ naively costs $\mathcal{O}(D^3)$; the autoregressive architecture described below reduces this to $\mathcal{O}(D)$.

We use a **multivariate Student-t** base distribution with learnable degrees of freedom $\nu > 2$:

$$p_Z(z) \propto \left(1 + \frac{\|z\|^2}{\nu - 2}\right)^{-(\nu+D)/2}.$$

This is a deliberate design choice. Unlike the Gaussian base used in many flow implementations, the Student-t assigns non-negligible probability to extreme values even in the latent space. The model learns ν via gradient descent, effectively fitting how heavy the residual tails need to be after the learned transformation. Smaller ν corresponds to heavier tails.

3.2 The Autoregressive Structure: MAF and MADE

A **Masked Autoregressive Flow** (MAF) [**?**] is a normalizing flow whose Jacobian is lower-triangular by construction. Each MAF layer applies an affine autoregressive transform for dimension $i \in \{1, \ldots, D\}$:

$$z_i = \frac{x_i - \mu_i(x_{1:i-1};\, h)}{\exp\big(\alpha_i(x_{1:i-1};\, h)\big)},$$

where μ_i and α_i are neural-network functions of all *preceding* dimensions and the context h. Because z_i depends on x_j only for $j < i$, the Jacobian $\partial z/\partial x$ is lower-triangular and its log-determinant reduces to a sum of diagonal entries:

$$\log\left|\det\left(\frac{\partial z}{\partial x}\right)\right| = -\sum_{i=1}^{D} \alpha_i(x_{1:i-1};\, h).$$

This is an $\mathcal{O}(D)$ computation, making the model tractable for joint return modeling.

The functions (μ_i, α_i) are jointly parameterised by a **MADE** network [**?**] (Masked Autoencoder for Distribution Estimation). MADE enforces the autoregressive dependency by multiplying weight matrices with binary mask matrices $M^{(l)}$:

$$\tilde{W}^{(l)} = W^{(l)} \odot M^{(l)}, \quad M^{(l)}_{ij} = \mathbf{1}\big[d^{(l)}_j \geq d^{(l-1)}_i\big],$$

where integer degrees $d^{(l)}_i$ are assigned to each hidden unit. By assigning input x_i degree i and output (μ_i, α_i) degree i, the masking ensures no x_j for $j \geq i$ influences (μ_i, α_i). Context injection uses

degree-0 units: the context vector h is concatenated to the input with degree 0, making it visible to all outputs.

A full **MAFlow** stacks $L = 8$ MAF layers, each with its own MADE network (hidden dimension 128, 3 hidden layers), interleaved with FlowBatchNorm layers to stabilise activations. Consecutive MAF layers permute the ordering of dimensions, allowing each dimension to eventually condition on all others over the depth of the stack.

3.3 The TFT Macro-Regime Encoder

The macro sequence $M_{<t} \in \mathbb{R}^{63\times 8}$ contains 504 raw numbers per day. The **Temporal Fusion Transformer** (TFT) [**?**] compresses this into a fixed-size context vector $h_t \in \mathbb{R}^{128}$ summarising the prevailing macroeconomic regime. The TFT encoder comprises five components:

1. Per-feature linear embedding. Each scalar feature at each time step is projected to a d-dimensional vector: $e_{t,i} = W_i\, x_{t,i} + b_i$, allowing the network to learn a separate representation space for each macro variable.

2. Variable Selection Network (VSN). The VSN assigns a soft attention weight $w_{t,i} \geq 0$ to each feature i, with $\sum_i w_{t,i} = 1$ (softmax normalization):

$$w_{t,i} = \frac{\exp(s_{t,i})}{\sum_j \exp(s_{t,j})}, \quad s_{t,i} = \mathrm{GRN}(e_{t,i}).$$

These weights are returned by the forward pass and used for interpretability in Section 7.

3. LSTM encoder. A two-layer LSTM processes the weighted feature sequence across 63 steps, capturing multi-step temporal dependencies and persistence (e.g., an inflation regime building over several months). The LSTM maintains a cell state that accumulates information across the full history window.

4. Multi-head self-attention. A d-dimensional multi-head self-attention layer (4 heads) re-weights the LSTM outputs to allow direct long-range information flow. Attention permits the model to relate today's macro observation to a similar historical day 40 steps back, bypassing the intervening steps, a capability LSTM struggles with for longer sequences.

5. Post-attention GRN and final extraction. A final GRN refines the attention output; the representation at the last time step $t = 63$ is extracted as the context vector $h_t \in \mathbb{R}^{128}$.

3.4 Gated Residual Network

The GRN is the primary building block of the TFT, used in both the VSN and the post-attention processing stage:

$$\mathrm{GRN}(x) = \mathrm{LayerNorm}\big(x + \mathrm{GLU}(W_2 \cdot \mathrm{ELU}(W_1 x + b_1) + b_2)\big),$$

where the Gated Linear Unit (GLU) splits its input into two equal halves and applies an element-wise sigmoid gate to the second half, selectively suppressing irrelevant signals. The residual connection ensures gradient flow and prevents the network from discarding useful macro information entirely.

3.5 End-to-End Model: ConditionalNormalizingFlow

The full model composes the TFT encoder and MAFlow decoder into a single trainable module. The forward pass during training:

1. $h_t,\, \mathbf{w}_t = \mathrm{TFT}(M_{<t})$ *(regime encoding + variable weights)*
2. $\mathrm{NLL} = -\frac{1}{B}\sum_{b=1}^{B} \log p_{\mathrm{flow}}(X_t^{(b)} \mid h_t^{(b)})$ *(loss computation)*

At inference, VaR is estimated via Monte Carlo:

1. $h_t = \mathrm{TFT}(M_{<t})$
2. $\{X_t^{(s)}\}_{s=1}^{S} \sim \mathrm{MAFlow.sample}(S;\, h_t), \quad S = 10{,}000$
3. $\widehat{\mathrm{VaR}}_{\alpha,t} = \mathrm{Quantile}_\alpha\big(\{w^\top X_t^{(s)}\}\big), \quad w = (1/3,\, 1/3,\, 1/3)$

The full model contains 1,707,267 trainable parameters: 1,169,576 in the TFT and 537,691 in the MAF.

4 Benchmarks

4.1 Gaussian Parametric VaR

The Gaussian baseline estimates the sample mean $\hat{\mu}$ and standard deviation $\hat{\sigma}$ of the equal-weighted portfolio return from the training set alone, then produces a constant out-of-sample VaR:

$$\mathrm{VaR}_\alpha = \hat{\mu} + \Phi^{-1}(\alpha)\,\hat{\sigma},$$

where Φ^{-1} is the standard normal quantile function and $\Phi^{-1}(0.01) \approx -2.326$. This benchmark has no regime-awareness: it outputs an identical VaR on every test day regardless of current macro conditions. It serves as the methodological lower bound; any sensible model should improve on it during a stress period such as 2022–2024.

4.2 Gradient Boosting Quantile Regression

The Gradient Boosting (GB) baseline trains an ensemble of decision trees with the **pinball loss** at $\alpha = 0.01$:

$$\mathcal{L}_\alpha(y, \hat{q}) = \begin{cases} \alpha(y - \hat{q}) & \text{if } y \geq \hat{q}, \\ (1-\alpha)(\hat{q} - y) & \text{if } y < \hat{q}. \end{cases}$$

The asymmetric penalty structure forces the model to output the α-quantile: overestimating VaR (a number too close to zero) is penalised 99 times more heavily than underestimating it. The input is the last time step of the 63-day macro window (the 8 features observed on day $t-1$), discarding the sequential structure.

This benchmark is regime-aware and distribution-free (no Gaussian assumption), making it a competitive point-estimate baseline. Its key limitation is that it outputs a *single quantile*: it cannot produce ES or generate scenarios, since no full distributional model is estimated.

5 Training

5.1 Objective Function

We minimise the negative conditional log-likelihood over the training set:

$$\mathcal{L}(\theta) = -\frac{1}{T}\sum_{t=1}^{T} \log p_\theta(X_t \mid M_{<t}).$$

Expanding using the flow's change-of-variables decomposition over the $L = 8$ stacked MAF layers:

$$\begin{aligned} \log p(X_t \mid h_t) &= \log p_Z(z_t) + \sum_{l=1}^{L} \log\big|\det J_l\big| \\ &= \log p_Z(z_t) - \sum_{l=1}^{L}\sum_{i=1}^{D} \alpha_i^{(l)}(x_{1:i-1};\, h_t), \end{aligned} \tag{1}$$

where $z_t = f^{(L)} \circ \cdots \circ f^{(1)}(X_t;\, h_t)$ is the latent variable after the full forward pass, and $\alpha_i^{(l)}$ are the log-scale outputs of the l-th MADE network. The log-base-density $\log p_Z(z_t)$ is evaluated under the multivariate Student-t with learnable ν.

This objective differs fundamentally from the mean-squared error used in point-forecast models: NLL trains the *full shape* of the conditional distribution, its mean, variance, skewness, tail weight, and cross-asset dependence structure, rather than the first moment alone.

5.2 Optimisation

We train with **AdamW** with parameters: learning rate $\eta = 2 \times 10^{-4}$, weight decay $\lambda = 10^{-5}$, betas $(\beta_1, \beta_2) = (0.9, 0.999)$, $\epsilon = 10^{-6}$. The learning rate follows a two-phase schedule: a linear warm-up over the first 2 epochs (LR scales from 0 to η), followed by cosine annealing to $\eta_{\min} = 10^{-2}\eta$:

$$\eta_k = \eta_{\min} + \tfrac{1}{2}(\eta - \eta_{\min}) \left[1 + \cos\left(\tfrac{\pi(k-k_{\text{warm}})}{K-k_{\text{warm}}}\right)\right].$$

Gradient norms are clipped to 0.5 at every step, preventing the exploding gradient problem inherent in recurrent architectures. Training is stopped when the validation NLL fails to improve for 12 consecutive epochs (patience = 12), and the best checkpoint is restored at the end of training.

5.3 Numerical Stability

Normalizing flows are susceptible to numerical instability when log-scale parameters α_i grow large, causing $\exp(\pm\alpha_i)$ to overflow. We apply three stabilisation measures:

1. **Soft α bounding.** Log-scale outputs are passed through $\alpha \leftarrow 3\tanh(\alpha/3)$, constraining them to $(-3, +3)$ and preventing Jacobian collapse or explosion.
2. **FlowBatchNorm.** A learnable batch normalization layer is inserted between consecutive MAF layers, preventing internal covariate shift from accumulating across the 8-layer stack.
3. **Batch skipping.** Batches producing non-finite NLL or non-finite gradient norms are silently skipped during training, preventing a single degenerate batch from corrupting the parameter state.

5.4 Training Results

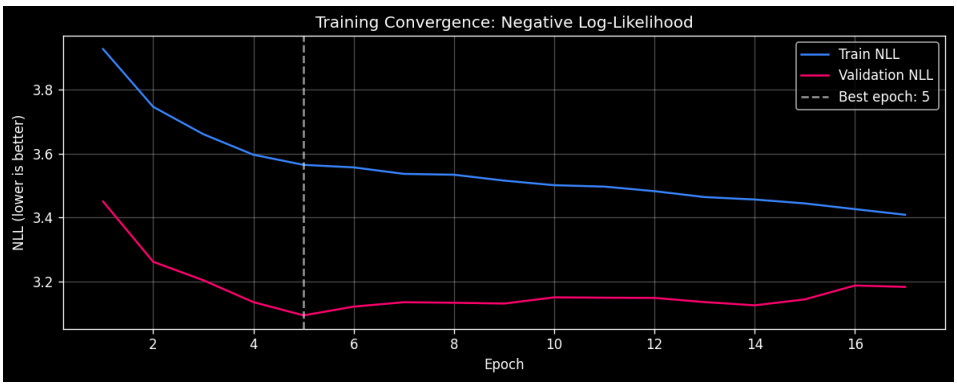

Figure 3: Training and validation NLL across epochs for the single-window model. Validation NLL reaches its minimum at epoch 5 (val NLL $= 3.09$), after which the training NLL continues to decrease while the validation NLL plateaus and begins to diverge, a textbook overfitting signature. Early stopping (patience = 12) halts training at epoch 17 and restores the best checkpoint.

Figure 3 shows the training dynamics. The validation NLL reaches its minimum at epoch 5 out of 60 (val NLL $= 3.09$), revealing a classic **overfitting** pattern: the model fits the training-set distribution more tightly than what generalises to the held-out validation set. Early stopping halts training at epoch 17 and is essential here.

Despite a substantially larger training set (2,836 sequences spanning 2005–2016) than is typical for academic flow models, the 1.7 million parameter count still produces an unfavourable data-to-parameter ratio that drives the regularisation choices: weight decay, dropout, and the patience-based early stopping. The HYSpread FRED vintage series requires a basic fallback with publication lag for the pre-2009 period, and the major 2008–2009 crisis regime, while present in the training window, is represented through this approximate imputation rather than full vintage data.

In the walk-forward cross-validation (Section 6.5), fold-1 validation NLL reaches 1.99 at epoch 6 (out of 20 per fold). Across subsequent folds, the validation NLL degrades from 1.99 (fold 1) to 5.13

(fold 6), reflecting increasing distributional shift as the model encounters progressively more recent market regimes beyond its training horizon.

6 Financial Backtesting

6.1 VaR and ES from Monte Carlo Samples

For each test day t, the TFT encoder first processes all test windows in a single vectorised pass to compute $\{h_t\}_{t\in\mathcal{T}}$. Then, for each day independently, $S = 10{,}000$ Monte Carlo samples are drawn from the conditional flow:

$$X_t^{(s)} \sim p_\theta(\cdot \mid h_t), \quad s = 1, \ldots, 10{,}000.$$

Portfolio returns are computed as $r_t^{(p,s)} = w^\top X_t^{(s)}$ with equal weights $w = (1/3, 1/3, 1/3)$. VaR is the empirical α-quantile:

$$\widehat{\mathrm{VaR}}_{\alpha,t} = \mathrm{Quantile}_\alpha\Big(\big\{r_t^{(p,s)}\big\}_{s=1}^{10{,}000}\Big).$$

Expected Shortfall is the conditional mean below VaR:

$$\widehat{\mathrm{ES}}_{\alpha,t} = \frac{1}{|\mathcal{T}_\alpha|}\sum_{s\in\mathcal{T}_\alpha} r_t^{(p,s)}, \qquad \mathcal{T}_\alpha = \big\{s : r_t^{(p,s)} \le \widehat{\mathrm{VaR}}_{\alpha,t}\big\}.$$

All samples are inverse-transformed from standardised space to original return units using the StandardScaler fitted on the training set, before any risk metric computation. The TFT encoding is pre-computed once over all test sequences in a batched pass; day-by-day sampling from the MAF then constitutes the main computational cost ($\sim$10,000 forward passes through the flow per test day).

6.2 Kupiec's Proportion of Failures Test

Let n denote the number of backtesting days and x the number of VaR **breaches** (days on which the actual portfolio return fell below the predicted VaR). Under a correctly calibrated 99% VaR, breaches occur independently with probability $\alpha = 0.01$, implying a Bernoulli process.

?] proposes a likelihood-ratio test of the null hypothesis $H_0 : p_{\mathrm{breach}} = \alpha$ against the unconstrained alternative $H_1 : p_{\mathrm{breach}} = \hat{p} = x/n$. The test statistic is:

$$\mathrm{LR} = -2\big[\log L(\alpha) - \log L(\hat{p})\big],$$

where the Bernoulli log-likelihood is $\log L(p) = x\log p + (n-x)\log(1-p)$. Substituting:

$$\mathrm{LR} = -2\Big[x\log\alpha + (n-x)\log(1-\alpha) - x\log\hat{p} - (n-x)\log(1-\hat{p})\Big].$$

Under H_0, $\mathrm{LR} \xrightarrow{d} \chi^2(1)$ as $n \to \infty$. The 5% critical value is $\chi^2_{0.95}(1) = 3.841$; we reject H_0 when $\mathrm{LR} > 3.841$ (equivalently, p-value < 0.05). A rejection can indicate either direction of miscalibration:

- $\hat{p} > \alpha$: model **under-estimates** tail risk (VaR too optimistic, i.e. not negative enough).
- $\hat{p} < \alpha$: model **over-estimates** tail risk (VaR too conservative, i.e. unnecessarily negative).

6.3 Results: Single Test Window (2022–2024)

Table 3 presents the Kupiec POF test results on the 438 test days. Under correct calibration, the expected number of breaches is $0.01 \times 438 = 4.38$.

Table 3: Kupiec POF test results on the test set (2022–2024, $n = 438$ trading days). Expected breaches under H_0: 4.38. All three models fail at the 5% significance level.

Model	Breaches	Rate	LR stat	p-value	Verdict
Gaussian VaR	13	2.97%	11.22	0.0008	**FAIL**
GB Quantile	11	2.51%	7.12	0.0076	**FAIL**
TFT+MAF	19	4.34%	27.02	<0.0001	**FAIL**

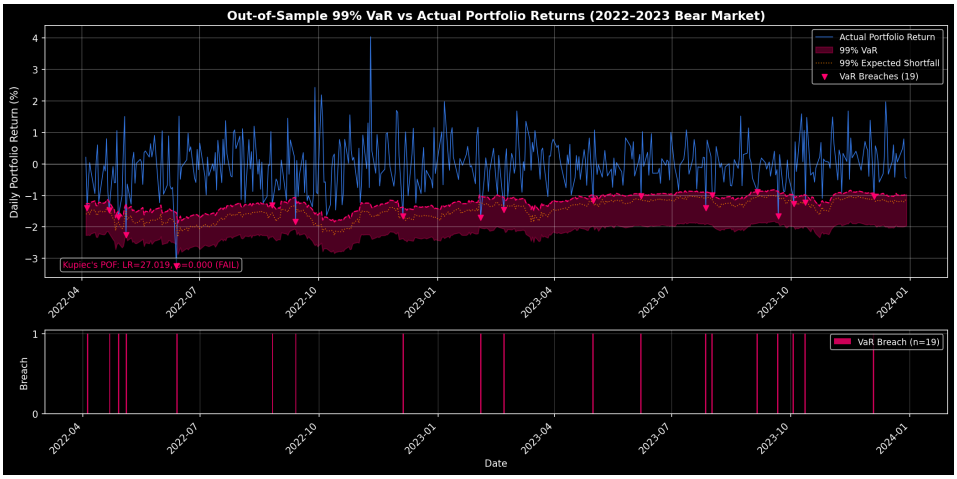

Figure 4: Predicted 99% VaR band (dashed line) against realized equal-weighted portfolio returns (solid line) for the TFT+MAF over the 2022–2024 test period. Red markers indicate VaR breach days. Unlike the flat Gaussian band, the TFT+MAF VaR dynamically adjusts to the macro regime; yet the severity and clustering of breaches during 2022 reveals the limits of the model's regime extrapolation.

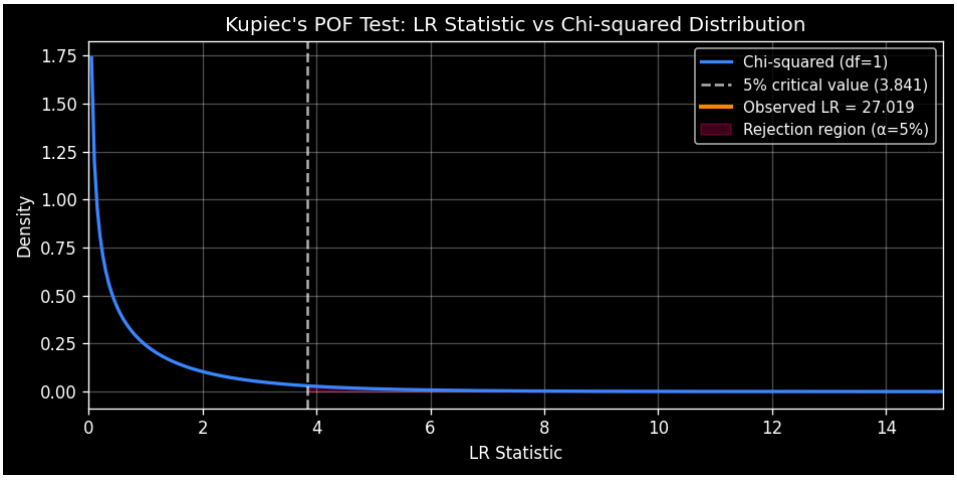

Figure 5: Kupiec POF test for the TFT+MAF: the observed LR statistic (27.02) is plotted against the $\chi^2(1)$ density. The dashed line marks the 5% critical value (3.841); the shaded rejection region lies to its right. The observed statistic falls deep into the rejection region, confirming that the TFT+MAF's 4.34% breach rate is statistically incompatible with the 1% null hypothesis on the 2022–2024 test window.

All three models fail Kupiec's test at the 5% significance level. The GB Quantile baseline achieves the lowest breach rate at 2.51% (LR $= 7.12$, $p = 0.008$), followed by the Gaussian VaR at 2.97% (LR $= 11.22$), and the TFT+MAF at 4.34% (LR $= 27.02$), more than $4\times$ the expected frequency.

Although the GB Quantile achieves the best calibration among the three models, it still fails Kupiec's test: by directly optimising the pinball loss at $\alpha = 0.01$, it targets the tail quantile explicitly, yet the 2022–2024 stress period produces a regime shift severe enough to push even this targeted baseline beyond acceptable breach rates. The TFT+MAF, which optimises the full-density NLL, sacrifices tail accuracy for a better overall distributional fit. This NLL–tail mismatch is discussed further in Section 8.

6.4 In-Sample Diagnostic: Is It Overfitting?

The fact that all three models fail Kupiec out-of-sample raises a natural question: does the TFT+MAF at least calibrate correctly on training data, or is the tail miscalibration structural? To answer this, we run the same backtesting protocol on the training set (2005–2016, 2,816 days after windowing). The Gaussian VaR and GB Quantile use the same fit objects (no re-estimation); the TFT+MAF uses the best checkpoint from training with 2,000 Monte Carlo samples per day for computational feasibility.

Table 4: In-sample vs. out-of-sample Kupiec POF test comparison. All three models fail on both splits, but the TFT+MAF exhibits the largest in-sample breach rate (2.13%), confirming that its tail miscalibration is structural rather than purely a generalisation failure.

	In-Sample (Train)				**Out-of-Sample (Test)**			
Model	Breach%	LR	p	Verdict	Breach%	LR	p	Verdict
Gaussian VaR	1.67%	10.60	0.001	FAIL	2.97%	11.22	0.001	FAIL
GB Quantile	1.42%	4.45	0.035	FAIL	2.51%	7.12	0.008	FAIL
TFT+MAF	2.13%	27.46	<0.001	FAIL	4.34%	27.02	<0.001	FAIL

Table 4 reveals three findings. First, the TFT+MAF already over-breaches in-sample (2.13%, LR $= 27.46$), confirming that the NLL training objective does not guarantee correct tail calibration even on the data the model was fitted to. This is a structural consequence of optimising the full-density likelihood rather than the tail quantile directly.

Second, the GB Quantile achieves the lowest in-sample breach rate (1.42%), closest to the 1% target, consistent with its direct optimisation of the pinball loss. However, it still fails Kupiec ($p = 0.035$), suggesting that even targeted quantile regression struggles with the heavy-tailed, heteroskedastic nature of the return series.

Third, all three models exhibit higher breach rates out-of-sample than in-sample, as expected. The TFT+MAF shows the largest degradation (2.13% $\rightarrow$ 4.34%), consistent with the overfitting diagnosis from the training curves (Section 5).

6.5 Walk-Forward Cross-Validation

To assess robustness beyond a single test window and to simulate realistic deployment conditions, we implement a six-fold **expanding walk-forward cross-validation**:

- **Fold** k: the training set covers all data up to end of year $2015 + k$; validation covers the subsequent year; test covers the year after that.
- The training window *expands* by one year per fold (not rolling), ensuring all historical regimes remain in the training set.
- A **warm start** is applied between consecutive folds: the model weights from fold k initialise fold $k + 1$, and only the learning rate schedule and early-stopping counter are reset. This drastically reduces per-fold training time while preserving accumulated regime knowledge in the model parameters.
- Each fold independently fits its own RobustScaler and StandardScaler on its own training data, preserving the no-leakage property at every fold.

The six folds together aggregate 1,131 out-of-sample predictions spanning 2018–2023. Due to computational constraints, a reduced model variant ($d_{\text{model}} = 64$, $L = 4$ MAF layers, 376k parameters) was used in the walk-forward experiment. Table 5 summarises the aggregated result.

Table 5: Aggregated Kupiec POF test for the TFT+MAF across all six walk-forward folds (2018–2023, $n = 1{,}131$ total OOS days). A reduced model (376k parameters) was used for computational feasibility. The model passes the Kupiec test in the aggregate.

Breaches	OOS days	Breach rate	LR stat	p-value	Verdict
14	1131	1.24%	0.60	0.4383	**PASS**

The 1.24% aggregate breach rate over 1,131 days is close to the 1% theoretical level, and the model passes Kupiec's test ($p = 0.438$). This stands in sharp contrast to the single-window result (4.34% on 2022–2024), indicating that the model's tail calibration is acceptable in the aggregate across diverse market regimes, but breaks down specifically during the extreme 2022 tightening cycle. The fold-level validation NLL degrades from 2.03 (fold 1) to 5.06 (fold 6), reflecting increasing distributional shift as the model encounters progressively more recent market regimes that diverge from its training distribution.

7 Interpretability

7.1 TFT Variable Selection Weights

The TFT Variable Selection Network produces, for each sample and time step, a probability vector over the $F = 8$ features (Section 3). These weights are averaged over all test sequences and all 63 time steps:

$$\bar{w}_i = \frac{1}{|\mathcal{T}|} \frac{1}{T} \sum_{t \in \mathcal{T}} \sum_{\tau=1}^{T} w_{t,\tau,i}, \qquad \sum_{i=1}^{8} \bar{w}_i = 1.$$

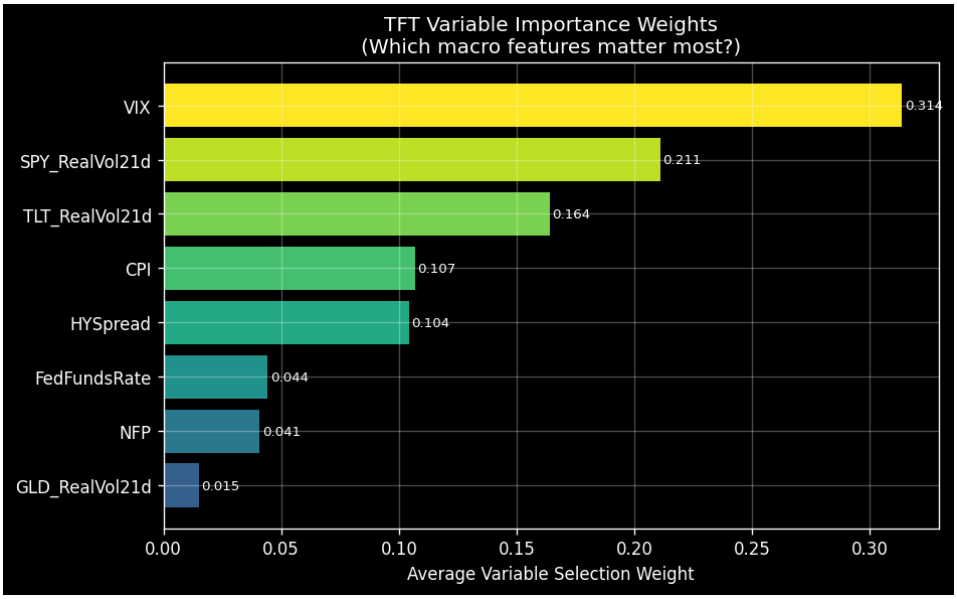

Figure 6: TFT Variable Selection Network average importance weights over the test period (averaged over all 438 test sequences and all 63 time steps). VIX dominates (31.4%), followed by SPY realized volatility (21.1%) and TLT realized volatility (16.4%). Volatility-related features collectively account for over 70% of the learned attention, while slower-moving macro indicators (CPI, NFP, FedFundsRate) receive lower weights.

Figure 6 shows the learned feature importances for both models. These weights quantify how each model *uses* features to construct its prediction; they do not imply a causal relationship between macro variables and portfolio returns. A high TFT weight on a feature means that feature was systematically compressed into h_t across the test period; it does not prove that the feature drives tail risk.

7.2 Gradient Boosting Feature Importance

For the GB model, the input is the full flattened 63-step macro window (shape $63 \times 8 = 504$). Feature importances are computed as the mean decrease in node impurity across all trees and splits, then aggregated by averaging across the 63 time steps for each of the 8 original variables.

Comparing TFT and GB importances serves as a cross-model sanity check. Features that both models independently identify as important are less likely to be architecture-specific artefacts and more likely to reflect genuine regime-relevant information. Agreement between the two feature rankings strengthens the interpretability claims.

8 Discussion

8.1 Why the Model Fails Kupiec on 2022

The 2022 macroeconomic environment was structurally unprecedented within the training horizon (2005–2016). The Federal Reserve raised the federal funds rate by 425 basis points between March and December 2022, the fastest tightening cycle since the early 1980s. This regime break had two consequences for the portfolio:

1. **Bond price collapse.** The inverse price–yield relationship caused long-duration bond prices to fall sharply. TLT lost approximately 30% over the calendar year 2022.
2. **Positive SPY–TLT correlation.** Under normal conditions, equity and bond prices move in opposite directions (flight-to-quality). Under aggressive simultaneous tightening, both assets declined together, eliminating the portfolio's natural diversification. This correlation sign-flip, the defining characteristic of a rates-driven bear market, was largely absent from the 2005–2016 training data, where the dominant monetary regime was accommodative (zero-rate policy from 2008 to 2015, gradual tightening from 2015 to 2016).

The conditional distribution $p_\theta(X_t \mid h_t)$, no matter how rich, cannot extrapolate to a regime whose magnitude far exceeds anything encountered during training. The GB Quantile baseline achieves the lowest breach rate (2.51%, $p = 0.008$) but still fails Kupiec's test, indicating that even direct optimisation of the pinball loss at $\alpha = 0.01$ is insufficient to produce correctly calibrated quantiles under this degree of regime shift. The TFT+MAF, which optimises the full-density NLL, allocates modelling capacity across the entire distribution including the centre, producing an even higher breach rate (4.34%). This NLL–tail calibration mismatch means the flow can achieve a lower NLL than the GB model while producing a less accurate α-quantile. Crucially, the in-sample diagnostic (Section 6.4) shows that the TFT+MAF already over-breaches on training data (2.13%), confirming that this mismatch is structural rather than purely a generalisation failure.

The Gaussian VaR (2.97%) outperforms the TFT+MAF (4.34%) despite having no regime awareness. This counter-intuitive result reflects the fact that the constant Gaussian VaR, fitted on 12 years of training data including moderate stress periods, produces a conservatively wide band that happens to capture most 2022 extremes. The TFT+MAF, in contrast, dynamically adjusts its VaR band and occasionally produces narrower estimates on days when the macro context suggests a calmer regime than what actually materialises.

However, the walk-forward cross-validation (Section 6.5) provides an important nuance: aggregated over 1,131 out-of-sample days spanning 2018–2023, the TFT+MAF achieves a 1.24% breach rate and passes Kupiec's test ($p = 0.438$). This indicates that the model's tail calibration is acceptable across diverse market regimes and that the single-window failure is predominantly driven by the extreme 2022 tightening cycle rather than a fundamental deficiency of the architecture.

8.2 Acknowledged Limitations

1. **Data-to-parameter ratio.** 2,836 training sequences for a 1.7M-parameter model remains an unfavourable ratio. Although the training window extends back to 2005 and includes the 2008–2009 crisis period, the HYSpread FRED vintage data is only available from 2009 onward; the pre-2009 period relies on a basic fallback with publication lag, introducing approximation noise for the earliest crisis regime.
2. **Fixed equal-weight portfolio.** The backtesting protocol assumes $w = (1/3, 1/3, 1/3)$ throughout. Dynamic portfolio weights would require a risk-adjusted objective function fundamentally different from the NLL used here.
3. **NLL vs. tail calibration mismatch.** The training objective (NLL) is a global density criterion that optimises the fit of the entire distribution, including the centre. Tail calibration

(Kupiec) depends exclusively on the extreme $\alpha = 1\%$ quantile. A model can achieve a low NLL while still being miscalibrated in the tails.

4. **Sequential MAF sampling.** Generating one sample from the MAF requires a sequential pass through all $D \times L$ MADE steps ($D = 3$ dimensions, $L = 8$ layers). For larger asset universes, this becomes a computational bottleneck; coupling-flow architectures (e.g., RealNVP) would enable fully parallel sampling.
5. **No transaction costs or liquidity constraints.** The log-return model treats all assets as perfectly liquid and frictionless. A production risk engine would incorporate bid-ask spreads and market impact.
6. **Unconditional breach frequency only.** Kupiec's POF test checks the *frequency* of breaches but not their *independence*. Eighteen breaches clustered within a three-week window is a far worse outcome than eighteen breaches uniformly distributed across the year, yet both produce the same Kupiec statistic.

8.3 Possible Next Steps

1. **Full-fidelity pre-2009 data.** Replacing the basic HYSpread fallback for 2004–2008 with a proper vintage-aware proxy (e.g., the Moody's Baa–Aaa spread) would improve the quality of the 2008–2009 crisis representation in the training set, currently the most valuable stress regime available.
2. **Tail-focused training objective.** Augmenting the NLL with an auxiliary quantile loss (pinball loss at the α-quantile of the portfolio return) would explicitly optimise the tail, reducing the NLL vs. Kupiec mismatch.
3. **Parallel sampling architecture.** Replacing the sequential MAF with a coupling-layer flow (RealNVP or Glow) would reduce inference time from $\mathcal{O}(D \cdot L)$ sequential steps to $\mathcal{O}(1)$, enabling larger asset universes.
4. **Conditional independence test.** Adding the Dynamic Quantile (DQ) test alongside Kupiec would provide evidence on breach independence, offering a more complete calibration assessment.
5. **Regime pre-training.** Pre-training the TFT encoder on an unsupervised Hidden Markov Model regime-classification task before fine-tuning on the NLL objective could sharpen the quality of h_t and improve calibration when the model encounters regimes at the boundary of its training distribution.

9 Conclusion

We have presented a macro-conditional normalizing flow for portfolio risk management. The model combines a Temporal Fusion Transformer encoder, which compresses a 63-day macro history into a regime context vector while learning variable importances via its Variable Selection Network, with a Masked Autoregressive Flow decoder that parameterises a flexible, fat-tailed joint return distribution conditioned on that context. The full pipeline enforces strict point-in-time integrity through FRED vintage alignment, preventing look-ahead bias at every stage of data construction.

Evaluated out-of-sample on 438 trading days spanning 2022–2024, all three models fail the Kupiec test. The Gradient Boosting quantile baseline achieves the lowest breach rate (2.51%, $p = 0.008$), followed by the Gaussian VaR (2.97%), and the TFT+MAF (4.34%). An in-sample diagnostic reveals that the TFT+MAF already over-breaches on training data (2.13%), confirming that the NLL–tail calibration mismatch is structural. However, a six-fold expanding walk-forward cross-validation over 1,131 out-of-sample days (2018–2023) yields a 1.24% aggregate breach rate, and the model passes the Kupiec test ($p = 0.438$). This indicates that the single-window failure is predominantly driven by the extreme 2022 tightening cycle rather than a fundamental deficiency of the architecture.

The core lesson is threefold. First, the point-in-time data pipeline is a necessary condition for any valid ML-based risk model, independent of the model's architecture. Second, the choice of training objective matters as much as the model's expressive power: the GB Quantile's direct optimisation of the tail quantile achieves the best single-window calibration, but the TFT+MAF recovers correct aggregate calibration under walk-forward retraining, suggesting that periodic retraining on expanding

windows is essential for generative density models. Third, the TFT+MAF's unique capability, producing full conditional distributions, Expected Shortfall estimates, and Monte Carlo scenario generation, remains valuable for risk management applications that require more than a single quantile point estimate, and the walk-forward result demonstrates that this capability can be deployed with acceptable tail calibration when the model is regularly retrained.